

gretl working papers

Bayesian Averaging of Classical Estimates (BACE) for **gretl**

Marcin Błażejowski

WSB University in Torun

marcin.blazejowski@wsb.torun.pl

Jacek Kwiatkowski

Nicolaus Copernicus University in Toruń

jkwiat@umk.pl

working paper #6

August 25, 2018

Abstract

This paper presents a software package that implements Bayesian Averaging of Classical Estimates **BACE** ver. 1.3 for the **gretl** (the GNU regression, econometrics and time-series library).

Project's main page:

https://sourceforge.net/projects/bace-for-gretl

Warning

This PDF is outdated and needs to be updated.

Contents

1 Introduction **2**

2 Bayesian Averaging of Classical Estimates **3**
2.1 XXX . . . 4
2.2 Jointness measures . . . 4

3 Implementation in gretl **6**
3.1 The core of the **BACE** code . . . 6
3.1.1 `BACE()` . . . 6
3.1.2 `BACE_GUI()` . . . 9
3.1.3 `BACE_Print()` . . . 10
3.1.4 `BACE_Plot()` . . . 10

4 Usage of the BACE package **10**
4.1 The GUI way . . . 10
4.2 The script way . . . 12
4.2.1 Script . . . 12
4.3 Output . . . 13

5 Empirical examples **15**
5.1 Results for UScrime dataset . . . 15
5.2 Results for birthwt dataset . . . 19

Changelog

1.3 Massive changes and improvements:

- Fixed bug in checking collinearity of independent variables – now correct estimator is used instead of hardwired OLS.
- (to doc) Correct model priors for truncated distributions.
- New MC3 strategy: taken directly from BMA package.
- New jointness measures.
- New `BACE()` bundle-based interface (default).
- Fix bug in `BACE_GUI()` in constant type selection.
- Fixed bug when „can be dropped" option is used.
- (to doc) New model types.
- (to doc) Updated docs.
- (to doc) Adding possibility for centering or standardizing variables.
- (do doc) New BIC versions.
- (do doc) Graphs added.
- (to doc) Overloading parameters in options bundle.
- Clean up the code.

1.2 Several changes:

- Some typos in output.
- Getting rid of calls to deprecated functions.
- New jointness measures.

1.1 Several changes:

- Documentation in PDF.
- Fix of several issues and bugs for model with only one explanatory variable and without constant.
- Change the option of the constant type from „random" to „can be dropped".

1.0 Initial version.

1 Introduction

In article Xavier Sala-i-Martin and Gernot Doppelhofer and Ronald I. Miller (2004) authors suggest a technique for averaging the parameter estimates – Bayesian Averaging of Classical Estimates (BACE), which enables to measure the importance of particular potential regressors. This approach is alternative to the familiar and

earlier-applied Bayesian Model Averaging (BMA) technique, from which it differs mainly because of its less restrictive *prior* assumptions. The authors showed that BACE weighting method may be understood as a limiting case of BMA approach when the *prior* information is dominated by the data. The averaging of the parameter estimates is performed across all possible combinations of linear regression models obtained by means of Ordinary Least Squares (OLS) with classical assumptions.

The BACE approach has several interesting features in relation to the techniques such as BMA or robustness-checking methods (reference). First of all, it is not necessary to make assumptions concerning *prior* distribution for parameters but only one simple assumption, on the expected model size.

An example of the application of the BACE technique for forecasting macroeconomic indicators can be found, among others, in the articles van Dijk (2004); Albis and Mapa (2014); Białowolski et al. (2014).

2 Bayesian Averaging of Classical Estimates

Consider simple regression model M_j:

$$y = X_j\beta_j + \epsilon \tag{1}$$

where y is $(Tx1)$ vector of observations, X_j is (Txk_j) matrix containing the set of independent variables and constant, β_j is a (k_jx1) vector of unknown parameters, ϵ is a $(Nx1)$ vector of errors which are assumed to be normally distributed i.e. $\epsilon \sim N(0_T, \sigma^2 I_T)$, $N(\mu, \Sigma)$ denotes a normal distribution with location μ and covariance Σ.

By applying Ordinary Least Squares (OLS) estimates (see Xavier Sala-i-Martin and Gernot Doppelhofer and Ronald I. Miller (2004)) we can calculate approximation of posterior probability of model M_j i.e. $\Pr(M_j|y)$ using the following formula: (add: Kass and Raftery (1995))

$$\Pr(M_j|y) \approx \frac{\Pr(M_j)T^{-k_j/2}SSE_j^{-T/2}}{\sum_{i=1}^{2^K}\Pr(M_i)T^{-k_i/2}SSE_i^{-T/2}}, \tag{2}$$

where SSE_j and SSE_i are the OLS sum of squared errors, 2^K denotes the total number of potential combinations of the K independent variables, k_j and k_i are the number of regression parameters β_j and β_i. Prior probabilities $\Pr(M_j)$ and $\Pr(M_i)$ of models M_j and M_i are binomially distributed, namely:

$$\Pr(M_j) = \theta^{k_i}(1-\theta)^{K-k_i}, \theta \in [0,1]. \tag{3}$$

Note that the Binomial distribution implies that we only need to specify a prior expected model size $\mathsf{E}(\Xi) = K\theta$ where $\mathsf{E}(\Xi) \in (0, K]$. If we define the value of $\mathsf{E}(\Xi)$ our BACE package will automatically produce a value of prior inclusion probability for all competitive models. In case of $\theta = 0.5$ prior expected model size is

equal to the average of number of potential regressors, model prior distribution is uniform $\left(\Pr(M_i) = 2^{-K}\right)$ and reflects a lack of previous knowledge about models.

Using BACE we can also easily evaluate the mean and variance of the posterior distribution of regression parameters β across the whole model space (see Leamer (1978)):

$$E(\beta|y) \approx \sum_{i=1}^{2^K} \Pr(M_i|y) E(\beta_i|y, M_i), \tag{4}$$

$$Var(\beta|y) \approx \sum_{i=1}^{2^K} \Pr(M_i|y) Var(\beta_i|y, M_i) + \sum_{i=1}^{2^K} \Pr(M_i|y) \left(E(\beta_i|y, M_i) - E(\beta|y)\right)^2, \tag{5}$$

where $E(\beta_i|y, M_i)$ and $Var(\beta_i|y, M_i)$ are the OLS estimates of β_i from model M_i.

Another useful characteristic popular in BACE approach is a Posterior Inclusion Probability (PIP). It is defined as the posterior probability that the variable x_i is relevant in explaining the dependent variable (see Leamer (1978); Mitchell and Beauchamp (1988)). In our case the PIP is calculated (using Monte Carlo integration) as the sum of the posterior model probabilities for all of the models including specific variable:

$$\Pr(\beta_i \neq 0|y) = \sum_{i=1}^{2^K} \Pr(M_i|y) \tag{6}$$

In many cases, despite of the interest of posterior characteristics it is valuable to predict future values of the dependent variable. In BACE approach we can carry out predictive inference across the whole model space. Given the matrix X_j^* containing explanatory variables for each of the out-of-sample data points we can predict inference on unobserved values of dependent variable as a weighted average of the individual model forecasts.

2.1 XXX

Add Beta-Binomial distribution (including truncated Atwood (1980); Tripathi et al. (1994))

2.2 Jointness measures

The main implementations of Bayesian model averaging are concerned with the selection of variables when model uncertainty is present. Another relevant issue which arises in this framework is to identify whether different sets of two variables x_i and x_j are substitutes, complements or neither over the model space. For that reason, Ley and Steel (2007) and Doppelhofer and Weeks (2009) define ex-post jointness measures of dependence between different sets of explanatory variables. Following Hofmarcher et al. (2018), let's define General contingency table for two binary variables:

Table 1: General contingency table for two binary variables.

	B	$\overline{B}$	Total
A	a	b	$a+b$
$\overline{A}$	c	d	$c+d$
Total	$a+c$	$b+d$	$a+b+c+d=N$

1. The jointness measure was proposed for the first time in Doppelhofer and Weeks (2006) as (`"Doppelhofer-Weeks Measure (DW1)"` in our package):

$$J_{DW1}(A,B) = \ln\left(\frac{\mathsf{P}(AB \mid y)}{\mathsf{P}(A \mid y)\mathsf{P}(B \mid y)}\right) \sim J_{DW1} = \ln\left(\frac{aN}{(a+b)(a+c)}\right). \quad (7)$$

2. An alternative jointness measure was proposed by Doppelhofer and Weeks (2009) as (`"Doppelhofer-Weeks Measure (DW2)"` in our package):

$$J_{DW2}(A,B) = \ln\left(\frac{\mathsf{P}(AB \mid y)\mathsf{P}(\overline{A}\overline{B} \mid y)}{\mathsf{P}(A\overline{B} \mid y)\mathsf{P}(\overline{A}B \mid y)}\right) \sim J_{DW2} = \ln\left(\frac{ad}{bc}\right). \quad (8)$$

3. Two jointness statistics were proposed in Ley and Steel (2007) as (`"Ley-Steel Measure (LS1)"` in our package):

$$J_{LS1}(A,B) = \frac{\mathsf{P}(AB \mid y)}{\mathsf{P}(A \mid y) + \mathsf{P}(B \mid y) - \mathsf{P}(AB \mid y)} \sim J_{LS1} = \frac{a}{a+b+c}, \quad (9)$$

4. and (`"Ley-Steel Measure (LS2)"` in our package):

$$J_{LS2}(A,B) = \frac{\mathsf{P}(AB \mid y)}{\mathsf{P}(A\overline{B} \mid y) + \mathsf{P}(\overline{A}B \mid y)} \sim J_{LS2} = \frac{a}{b+c}. \quad (10)$$

5. Another jointness measure was proposed in **?** as (`"Strachan Measure"` in our package):

$$J_{St}(A,B) = \mathsf{P}(A \mid y)\mathsf{P}(B \mid y) \ln\left(\frac{\mathsf{P}(AB \mid y)}{\mathsf{P}(A\overline{B} \mid y)\mathsf{P}(\overline{A}B \mid y)}\right) \sim J_{ST} = \frac{a+b}{N}\frac{a+c}{N} \ln\left(\frac{aN}{bc}\right). \quad (11)$$

6. Next jointness measure is Yule's Q defined as (`"Yule's Q Measure (YQ)"` in our package):

$$J_{YQ}(A,B) = \frac{\mathsf{P}(AB)\mathsf{P}(\overline{A}\overline{B}) - \mathsf{P}(A\overline{B})\mathsf{P}(\overline{A}B)}{\mathsf{P}(AB)\mathsf{P}(\overline{A}\overline{B}) + \mathsf{P}(A\overline{B})\mathsf{P}(\overline{A}B)} \sim J_{YQ} = \frac{ad-bc}{ad+bc}. \quad (12)$$

7. The last jointness measure (the default) is modified Yule's Q defined as (`"Modified Yule's Q Measure (YQM)"` in our package):

$$J_{YQ} = \frac{\left(a+\frac{1}{2}\right)\left(d+\frac{1}{2}\right) - \left(b+\frac{1}{2}\right)\left(c+\frac{1}{2}\right)}{\left(a+\frac{1}{2}\right)\left(d+\frac{1}{2}\right) + \left(b+\frac{1}{2}\right)\left(c+\frac{1}{2}\right) - \frac{1}{2}}. \quad (13)$$

According to Doppelhofer and Weeks (2009), we use the classification of jointness among the variables given in Table 2.

Table 2: Classification of strength of jointness measures.

Evidence	Jointness statistics
strong substitutes	$J_{LS}, J_{DW} \leq -2$
significant substitutes	$-2 < J_{LS}, J_{DW} \leq -1$
not significantly related	$-1 < J_{LS}, J_{DW} < 1$
significant complements	$1 \geq J_{LS}, J_{DW} < 2$
strong complements	$J_{LS}, J_{DW} \geq 2$

3 Implementation in gretl

In this section, we describe the code of the **BACE** package. First, we will characterize our code and the use of both: the command line and the GUI interfaces, then we will present some examples of using the **BACE**. Finally, we will show the outputs that are returned.

3.1 The core of the BACE code

The **BACE** package consists of 4 public and 14 private functions. The name of each function starts with the `BACE_` prefix. The public functions are:

1. `BACE()` see 3.1.1
2. `BACE_GUI()` see 3.1.2
3. `BACE_Print()` see 3.1.3
4. `BACE_Plot()` see 3.1.4

3.1.1 `BACE()`

This is a core package function which runs and controls the main loop. As a consequence, it is a prime interface to the **BACE** package. The `BACE()` function returns a bundle with all information needed to print and/or plot results anytime. One can print the results by calling `BACE_Print()` or plot graphs by calling `BACE_Plot()`. The signature of `BACE()` function is as follows:

```
function bundle BACE (series Y "Dependent variable",
  list X_list "List of indenpendent variables (X)",
  bundle Options[null])
```

The first two options are mandatory, while the last one is an optional collection of additional arguments. Those arguments are defined with the following default values:

```
scalar Options["Y_AR_p"] = 0
scalar Options["with_const"] = 1
```

```
scalar Options["model_prior"] = 2
scalar Options["avg_model_size"] = avg_model_size_tmp
scalar Options["alpha"] = 0.6
scalar Options["Nrank"] = xmin(2^(k+Y_AR_p_tmp-1), 40)
scalar Options["do_joint"] = 0
scalar Options["Npredict"] = 0
scalar Options["Nrep"] = Nrep
scalar Options["burn"] = 10
scalar Options["verbosity"] = 2
scalar Options["show_progress"] = do_show_progress
scalar Options["do_plot"] = do_plot_tmp
string Options["model_type"] = "linear"
scalar Options["data_trans"] = 0
scalar Options["bic_type"] = 1
scalar Options["hyperparameter"] = 0.5 # gamma for EBIC
scalar Options["binary_threshold"] = 0.5
scalar Options["low"] = 0
scalar Options["high"] = NA
scalar Options["_gui"] = 0
```

where k stands for the number of independent variables, `scalar avg_model_size` is calculated according to the following formula:

$$\texttt{avg_model_size} = \begin{cases} \texttt{(k + Y_AR_p_tmp)/2} & \text{if} \quad \texttt{with_const != 2}, \\ \texttt{(k + Y_AR_p_tmp + 1)/2} & \text{elsewhere}, \end{cases} \tag{14}$$

`scalar Nrep` takes the following values:

$$\texttt{Nrep} = \begin{cases} 10^4 & \text{if} \quad k \leq 15, \\ 10^5 & \text{if} \quad k \in [16; 19], \\ 10^6 & \text{if} \quad k \geq 20, \end{cases} \tag{15}$$

`scalar show_progress` takes the following values:

$$\texttt{show_progress} = \begin{cases} 1 & \text{if} \quad \texttt{\$sysinfo.gui_mode}, \\ 0 & \text{elsewhere}, \end{cases} \tag{16}$$

and `scalar do_plot` takes the following values:

$$\texttt{do_plot} = \begin{cases} 1 & \text{if} \quad \texttt{\$sysinfo.gui_mode}, \\ 0 & \text{elsewhere}. \end{cases} \tag{17}$$

Most of the optional parameters listed above are of numerical type and they may be gathered into following groups according to what kind of numbers they can take[1]:

1. Integers: `Y_AR_p`, `with_const`, `model_prior`, `Nrank`, `do_joint`, `Npredict`, `Nrep`, `burn`, `verbosity`, `data_trans`, `bic_type`.

[1] Note, that the `scalar _gui` is for internal use only and must not be modified by user at any rate.

2. Booleans: `show_progress`, `do_plot`.

3. Scalars: `avg_model_size`, `alpha`, `binary_threshold`.

Optional parameters control the following package options:

1. **Y_AR_p**: autoregression order of variable Y.

2. **with_const**: whether we include constant term in each estimated model?; possible values are: 0—without constant, 1—constant is always included, 2—constant may be dropped.

3. **model_prior**: model prior distribution to use; possible values are: 0—none, 1—Binomial, 2—Beta-Binomial.

4. **avg_model_size**: expected average model size for model prior.

5. **alpha**: significance level for the initial model.

6. **Nrank**: number of top ranked models.

7. **do_joint**: do we perform jointness analysis?; possible values are: 0—none, 1—Doppelhofer-Weeks Measure (DW1), 2—Doppelhofer-Weeks Measure (DW2), 3—Ley-Steel Measure (LS1), 4—Ley-Steel Measure (LS2), 5—Strachan Measure, 6—Yule's Q Measure (YQ), 7—Modified Yule's Q Measure (YQM).

8. **Npredict**: number of out-of-sample forecasts. Note, that using this option makes **BACE** to perform an analysis on subsample.

9. **Nrep**: total number of Monte Carlo iterations (model draws).

10. **burn**: percentage of burn-in draws in MC^3.

11. **verbosity**: verbosity of the output: 1—basic results, 2—full results.

12. **show_progress**: do we show overall progress?

13. **do_plot**: do we produce graphs?

14. **model_type**: model type we should apply to the data; possible options are: 1—linear, 2—logit, 3—probit, 4—poisson.

15. **data_trans**: shall the **BACE** transform data before analysis?; possible options are: 0—none, 1—centering, 2—standarizing.

16. **bic_type**: Bayesian information criterion version which should be used; possible values are: 1—bic, 2—hbic, 3—ibic, 4—ebic.

17. **hyperparameter**: hyperparameter for prior setup; for now it is interpreted as γ for ebic.

18. **binary_threshold**: applicable only for binary-dependent models; this value indicates the lowest probability for which $\hat{y}_i$ should take 1.
19. **low**: applicable only for `tobit` model; this option controls tobit's `--llimit` flag for setting left bound in censoring.
20. **high**: applicable only for `tobit` model; this option controls tobit's `--rlimit` flag for setting right bound in censoring.

Some of the above-mentioned parameters are over-loadable, which means that their value may be passed either via an integer or via string (case insensitive). Those parameters (together with value-string mappings) are:

bic_type : 'bic'=1, 'hbic'=2, 'ibic'=3, 'ebic'=4;

model_prior : 'none'=0, 'binomial'=1, 'betabinomial'=2 (or 'beta-binomial'=2);

with_const : 'never'=0, 'always'=1, 'canbedropped'=2 (or 'can be dropped'=2);

do_joint : 'none'=0, 'dw1'=1, 'dw2'=2, 'ls1'=3, 'ls2'=4, 'sm'=5, 'yq'=6, 'myq'=7;

data_trans : 'none'=0, 'centering'=1, 'standarizing'=2;

model_type : 'linear'=1, 'logit'=2, 'probit'=3, 'poisson'=4;

3.1.2 BACE_GUI()

The `BACE_GUI()` function is an interface to the `BACE()` its signature is as follows:

```
function bundle BACE_GUI (
  int model_type[1:5:1] "Model type" {"logit (including ordered)",
        "probit (including ordered)", "poisson", "tobit",
        "linear"},
  series Y "Dependent variable",
  int Y_AR_p[0::0] "Y lags (time-series only)",
  list X_list[null] "List of independent variables (X)",
  int with_const[0:2:1] "Constant in model" {"Never", "Always",
        "Can be dropped"},
  int model_prior[0:2:1] "Model prior" {"none", "Binomial",
        "Beta-Binomial"},
  int bic_type[1:4:1] "Bayesian information criterion" {"BIC",
        "HBIC", "IBIC", "EBIC"},
  int do_joint[0:7:0] "Jointness analysis" {"None",
        "Doppelhofer-Weeks Measure (DW1)", "Doppelhofer-Weeks
        Measure (DW2)", "Ley-Steel Measure (LS1)", "Ley-Steel
        Measure (LS2)", "Strachan Measure (SM)", "Yule's Q
        Measure (YQ)", "Modified Yule's Q Measure (MYQ)"},
  int Npredict[0::0] "Number of out-of-sample forecasts",
  int verbosity[1:2:2] "Verbosity",
  bool show_progress[1] "Show overall progress",
  bool do_plot[1] "Plot graphs?")
```

Note, that although BACE_GUI() is designed to be a wrapper to BACE() it does not allow for setting all possible parameters. This is partially because some of them are mutually exclusive, partially because we wanted to design GUI interface which is not very complicated.

3.1.3 BACE_Print()

BACE_Print() is a function for printing results of analysis performed by **BACE** package and its signature is as follows:

```
function void BACE_Print (bundle *MODEL)
```

This function takes as an argument reference to the bundle MODEL with **BACE** results and prints is according to verbosity settings.

3.1.4 BACE_Plot()

BACE_Plot() is a function for producing graphs and its signature is as follows:

```
function void BACE_Plot (const bundle *MODEL)
```

This function takes as an argument reference to the MODEL bundle with **BACE** results and plots graphs according to do_plot settings. In non-GUI mode (when $sysinfo.gui_mode == 0), plots are generated as scripts for gnuplot program and saved on a disk. Names of generated files have their base set to strftime($now[1], "%Y%m%d%H%M%S").

4 Usage of the BACE package

4.1 The GUI way

Once you start gretl, you must open a data file and then you can load the relevant BACE package from the gretl server. In the main window, go to File → Function packages → On server heading. By selecting BACE, you will download the package on your local machine. Next, go to File → Function packages → On local machine and by selecting BACE from the list open a window similar to the one shown in Figure 1:

According to Figure 1, we can specify the following entries in the GUI BACE window

- **Model type** – What kind of modeling should be should be performed? [required]
- **Dependent variable** – The dependent variable as series [required].
- **Y lags** – number of lags for dependent variable as integer (for time-series only).

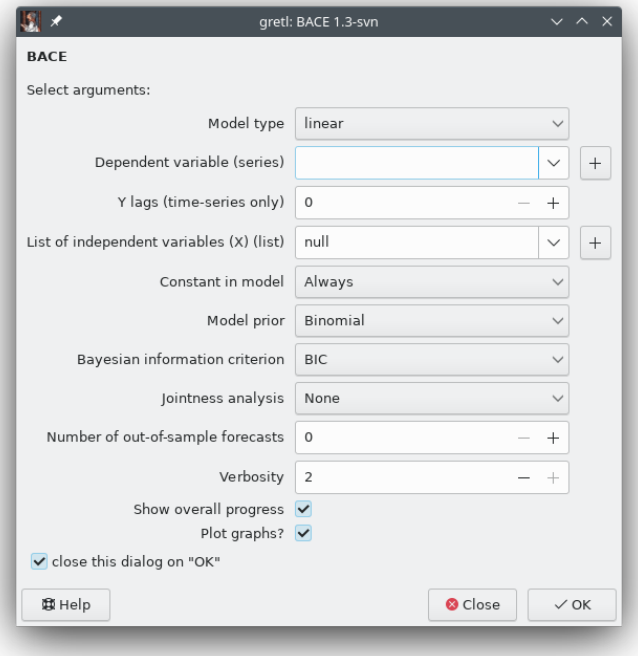


Figure 1: Main window for `BACE`.

- **List of independent variables** – List of all independent variables (including lagged series) as list [may be null].
- **Constant in model** – Do we include intercept? [required]
  1. Never – all models without constant,
  2. Always – all models with constant,
  3. Can be dropped – constant may be removed from or added to any model.
- **Model prior** – Do we impose a prior on model space? [required]:
  1. none,
  2. Binomial – binomial distribution,
  3. Beta-Binomial – beta-binomial distribution.
- **Bayesian information criterion** – Version of BIC to be used [required].
  1. BIC,
  2. HBIC,
  3. IBIC,
  4. EBIC.

- **Jointness analysis** – If `None` (the default), the jointness analysis is omitted. Alternatively, one can choose one of the 7 available jointness measures.
- **Number of out-of-sample forecasts** – Defines the total number of out-of-sample forecasts of the dependent variable.
- **Verbosity** – An integer value of 1 or 2; the default is 2, so a more detailed description of the analysis is provided (initial model, speed of convergence, estimation results for top ranked models).
- **Show overall progress** – The progress bar is printed.
- **Plot graphs?** – Shall we produce graphs?

4.2 The script way

4.2.1 Script

The `BACE` package can also be used inside `Hansl` scripts. A very minimal code could be as follows:

```
open greene9_1.gdt
include BACE.gfn

list greene = capital labor nfirm

BACE(valadd, greene)
```

The above example consists of three blocks. The first block is just opening of the so called `greene9_1` dataset, which is bundled in every standard **gretl** installation. This dataset contains cross-sectional data on manufacturing of transportation equipment presented as Table 9.1 in Greene (1999)[2]. We also set the seed in order to have exact reproduction of the results.

The second block is the definition of the `greene` list which contains three independent variables available in the `greene9_1` dataset.

The third block contains the definition for Bayesian Averaging of Classical Estimates analysis with default settings described on page 6. On contrary, the corresponding "big call", i.e. setting all possible parameters manually, would be as follows:

```
bundle opts = defbundle("model_prior", 2, "avg_model_size", 1.5, \
  "alpha", 0.6, "Nrank", 4, "do_joint", 0, "Npredict", 0, \
  "Nrep", 1000, "burn", 10, "verbosity", 2, "show_progress", 1, \
  "do_plot", 1, "model_type", "linear", "const_type", 1, \
  "bic_type", 1, "data_trans", 0)
BACE(valadd, greene, opts)
```

or alternatively with the new syntax for creating bundle on-the-fly:

[2]This dataset is also available at http://people.stern.nyu.edu/wgreene/Text/Edition7/TableF7-2.txt.

```
BACE(valadd, greene, _(model_prior=2, avg_model_size=1.5,\
  alpha=0.6, Nrank=4, do_joint=0, Npredict=0, Nrep=1000,\
  burn=10, verbosity=2, show_progress=1, do_plot=1,\
  model_type="linear", const_type=1, bic_type=1, data_trans=0))
```

Since some parameters in option bundle are over-loadable (see 3.1.1), the equivalent call looks like this:

```
BACE(valadd, greene, _(model_prior="beta-binomial",\
  avg_model_size=1.5, alpha=0.6, Nrank=4, do_joint="none",\
  Npredict=0, Nrep=1000, burn=10, verbosity=2, show_progress=1,\
  do_plot=1, model_type="linear", const_type="always",\
  bic_type="bic", data_trans="none"))
```

Suppose we want to run the same analysis but following Occam's razor principle. The simplest way is setting the prior average model size to 1. We also want to have "random" constant, i.e. intercept can be dropped in any model. We also want to save results of the BACE analysis in the bundle `res`, the code should be as follows:

```
res = BACE(valadd, greene, _(avg_model_size=1.1, with_const=2))
```

Suppose we want to calculate Doppelhofer-Weeks Measure (DW1) instead of already computed Modified Yule's Q Measure (MYQ). Since the type of jointness measure is controlled by `do_join` field in the result bundle, the only thing we have to do is to set the value of `do_join` to 2 and print BACE results again. The code should be as follows:

```
res.do_joint = 2
BACE_Print(&res)
```

4.3 Output

Outdated!!!

If you select the appropriate entries in the GUI `BACE` window, our package returns: Posterior Inclusion Probabilities (PIP), the averages of the posterior means and standard deviations of parameters ("Mean" and "Std.Dev.", respectively) and the posterior mean and standard deviation of each coefficient conditional on the variable being included in the model ("Cond.Mean" and "Cond.Std.Dev."). Furthermore, the package also returns the predictive results for the dependent variable ("Mean" and "Std.Dev.").

Let us consider the data used in previous example (`australia.gdt`). This dataset contains the Australian macro data used in Johansen (1995) and originally supplied by Tony Hall. The variables used in the text are:

- lpau – log of Australian CPI (dependent variable),
- lpus – log of US CPI,
- le – log of exchange rate (price of US dollars in Aus. dollars),

- iau2 – 5-year Treasury bond rate, Australia,
- ius2 – 5-year Treasury bond rate, USA.

Suppose we want to perform BACE analysis up to $ADL(2,2)$ specification, i.e. all possible combinations from $ADL(0,0)$ to $ADL(2,2)$. The example script would be as follows:

```
set verbose off
include BACE.gfn
open australia.gdt
genr time

# We create lagged values
adl_lags = 2
lags(adl_lags, ius2)
lags(adl_lags, iau2)
lags(adl_lags, e2)
lags(adl_lags, lpus)
lags(adl_lags, le)

# We define list with all X-es (including lags)
list X_list = time ius2 ius2_* iau2 iau2_* e2 e2_* lpus lpus_* le le_*

set seed 1000000
scalar y_lags = 2
scalar with_const = 2
scalar model_prior = 1
scalar prior_avg_model_size = (nelem(X_list) + y_lags + (with_const > 0)) / 2
scalar alpha = 0.6
scalar top_ranked_model = 4
scalar joint = 0
scalar forecasts = 4
scalar Nrep = 20000
scalar Nburn = 10
scalar verbosity = 2
scalar progress = 0

RES = BACE_GUI(lpau, y_lags, X_list, with_const, model_prior, prior_avg_model_s
alpha, top_ranked_model, joint, forecasts, Nrep, Nburn, verbosity, progress)
```

In the above analysis all variables (including dependent variable $lpau_t$) are lagged by 2 periods. Additionally we compute 4 forecasts:

```
Posterior moments (unconditional and conditional on inclusion):
PIP         Mean     Std.Dev.    Cond.Mean Cond.Std.Dev
lpau_1      1.000000     0.894595     0.091656     0.894595     0.09165
ius2      0.996500    -0.418560     0.102196    -0.420030     0.099314
```

```
lpus_2      0.790611    -0.303217     0.212152    -0.383523     0.16164
lpus_1      0.780500     0.356760     0.260342     0.457092     0.20243
iau2      0.752222     0.224487     0.161678     0.298432     0.112617
le_2      0.506944    -0.027670     0.046149    -0.054582     0.052271
const      0.392778    -0.074241     0.125663    -0.189015     0.136050
e2_2      0.335667     0.002317     0.033496     0.006903     0.057540
lpus      0.296222     0.037894     0.107134     0.127924     0.165014
le_1      0.282500    -0.023085     0.062683    -0.081716     0.095485
lpau_2      0.253056     0.037914     0.087083     0.149825     0.11489
e2_1      0.245222     0.015275     0.052532     0.062292     0.091240
time      0.233778     0.000053     0.000269     0.000226     0.000520
iau2_1      0.231389     0.035210     0.090726     0.152166     0.13332
le      0.211667     0.010063     0.042201     0.047542     0.081437
e2      0.176222    -0.005057     0.034623    -0.028698     0.078258
ius2_1      0.175222     0.021296     0.072659     0.121539     0.13396
ius2_2      0.140111    -0.009023     0.048048    -0.064400     0.11362
iau2_2      0.132000     0.005342     0.042150     0.040469     0.10971

---------------------------------

Posterior probability of models:
Model 1:        0.024278
Model 2:        0.020556
Model 3:        0.012889
Model 4:        0.009278
Total probability of the models in ranking (numerical): 0.067000

Correlation coefficient between the analytical
and numerical probabilities of the above models: 0.932557

---------------------------------

Predictive results:
lpau           Mean       Std.Dev.
1990:2        5.334649       5.340552       0.007950
1990:3        5.341856       5.356902       0.008400
1990:4        5.368310       5.378660       0.010094
1991:1        5.366443       5.395817       0.013705
```

5 Empirical examples

5.1 Results for UScrime dataset

Variable	BMA_RIC	BACE1_RIC	BACE2_RIC	BMA_UIP	BACE1_UIP	BACE2_UIP	reParMA	bicreg	iBMA.bicreg
M	0.7535	0.7237	0.7094	0.8493	0.8857	0.8777	0.7421	0.93	0.946
So	0.1430	0.1191	0.0423	0.2352	0.2082	0.2346	0.1458	0.209	0.155
Ed	0.9424	0.9322	0.7719	0.9756	0.9888	0.9801	0.9480	0.997	0.999
Po1	0.6141	0.6355	0.7132	0.6308	0.6505	0.7005	0.6390	0.693	0.707
Po2	0.4290	0.3951	0.3939	0.4551	0.4234	0.3945	0.4014	0.387	0.355
LF	0.0842	0.0721	0.1519	0.1604	0.1392	0.1805	0.0853	0.141	nd
MF	0.0947	0.0760	0.3010	0.1572	0.1350	0.1989	0.0973	0.147	0.124
Pop	0.2316	0.1910	0.0234	0.3300	0.3270	0.3758	0.2242	0.351	0.342
NW	0.4989	0.4529	0.0563	0.6773	0.7405	0.7308	0.5126	0.798	0.842
U1	0.1104	0.0887	0.0787	0.2054	0.1958	0.2287	0.1189	0.212	0.191
U2	0.4644	0.3999	0.2269	0.6051	0.6584	0.6720	0.4403	0.715	0.741
GDP	0.1850	0.1452	0.2252	0.3241	0.3149	0.3677	0.1784	0.357	0.35
Ineq	0.9943	0.9929	0.9909	0.9970	0.9997	0.9975	0.9973	1	1
Prob	0.7912	0.7384	0.2061	0.9027	0.9343	0.9086	0.7730	0.963	0.973
Time	0.1842	0.1488	0.0429	0.3238	0.3553	0.3874	0.1815	0.41	0.42

Table 3: PIPs obtained for UScrime datasets in different packages and simulation scenarios

Variable	BMA_RIC	BACE1_RIC	BACE2_RIC	BMA_UIP	BACE1_UIP	BACE2_UIP	rParMA	bicreg	iBMA.bicreg
GDPsh560	0.9995	0.9984	0.9991	1	1.0000	1.0000	0.9989	1	1
Confuncious	0.9885	0.9860	0.9700	0.9987	1.0000	0.9987	0.9887	1	1
Life_Exp	0.9324	0.9267	0.9333	0.9876	0.9990	0.9883	0.9306	1	1
Equip_Inv	0.9244	0.9316	0.9282	0.9227	0.9844	0.9575	0.9202	1	1
SubSahara	0.7263	0.7183	0.6920	0.9495	0.9933	0.9508	0.7350	1	1
Muslim	0.6573	0.6190	0.6313	0.5123	0.5488	0.5404	0.6350	0.697	0.598
Rule_of_Law	0.4818	0.4462	0.4497	0.7578	0.9181	0.8253	0.4964	0.998	1
Yrs_Open	0.5229	0.5344	0.5830	0.2458	0.1732	0.2241	0.5047	na	na
Eco_Org	0.4447	0.4096	0.3362	0.6383	0.5820	0.5858	0.4599	0.318	0.335
Protestants	0.4434	0.4316	0.3760	0.5698	0.4622	0.4878	0.4597	na	0.079
NEquip_Inv	0.4221	0.3878	0.3241	0.729	0.8306	0.7386	0.4330	0.745	0.868
Mining	0.4622	0.4259	0.4039	0.8939	0.9692	0.8773	0.4576	0.99	0.994
LatAmerica	0.2034	0.2007	0.2267	0.6335	0.8802	0.7059	0.2146	0.957	1
PrSc_Enroll	0.2238	0.1892	0.3112	0.5011	0.7710	0.5976	0.2011	0.983	0.918
Buddha	0.2047	0.1660	0.1716	0.3526	0.4291	0.3630	0.1983	0.409	0.363
Bl_Mkt_Pm	0.1803	0.1502	0.1532	0.6437	0.7336	0.6175	0.1806	0.745	0.698
Catholic	0.1270	0.1059	0.1271	0.2353	0.2172	0.1923	0.1352	0.145	0.058
Civl_Lib	0.1328	0.1121	0.1992	0.4488	0.6711	0.5176	0.1298	0.991	0.876
Hindu	0.1205	0.0955	0.0968	0.8666	0.9855	0.8072	0.1261	0.999	1
Pr_Exports	0.1028	0.0839	0.1346	0.1793	0.1770	0.1527	0.0952	0.122	0.021
Pol_Rights	0.0983	0.0815	0.1380	0.3430	0.3557	0.2922	0.0951	na	0.11
R_FEX_Dist	0.0834	0.0757	0.0875	0.2020	0.2506	0.1857	0.0819	0.207	0.081
Age	0.0900	0.0779	0.0905	0.3146	0.3366	0.2704	0.0866	na	0.105
War_Dummy	0.0766	0.0672	0.0884	0.2637	0.3090	0.2333	0.0763	na	0.119

Variable	BMA_RIC	BACE1_RIC	BACE2_RIC	BMA_UIP	BACE1_UIP	BACE2_UIP	rParMA	bicreg	iBMA.bicreg
English_pct	0.0699	0.0566	0.0700	0.369	0.6126	0.4507	0.0728	0.747	0.87
Foreign_pct	0.0664	0.0606	0.0778	0.1495	0.1618	0.1261	0.0657	0.105	0.02
Lab_Force	0.0744	0.0548	0.0746	0.7963	0.9761	0.7701	0.0745	0.999	1
Spanish_Col	0.0555	0.0451	0.0821	0.4541	0.8142	0.5817	0.0555	0.96	0.987
EthnoL_Frac	0.0575	0.0447	0.0686	0.6828	0.9414	0.7153	0.0545	0.999	1
std_BMP_	0.0503	0.0433	0.0501	0.1250	0.1459	0.1126	0.0484	na	na
French_Col	0.0502	0.0422	0.0604	0.4128	0.7813	0.5534	0.0520	0.96	0.98
Abs_Lat	0.0426	0.0359	0.0679	0.1571	0.1595	0.1251	0.0443	na	na
Work_Pop	0.0431	0.0378	0.0635	0.1327	0.1470	0.1110	0.043	na	0.011
High_Enroll	0.0417	0.0353	0.0629	0.6881	0.9555	0.7237	0.0420	0.999	1
Pop_g	0.0368	0.0322	0.0704	0.1473	0.1736	0.1267	0.0384	na	0.036
Brit_Col	0.0378	0.0316	0.0423	0.3288	0.7091	0.4912	0.0362	0.938	0.98
Outwar_Or	0.0352	0.0288	0.0375	0.3312	0.6846	0.4697	0.0370	0.83	0.86
Jewish	0.0361	0.0311	0.0424	0.1225	0.1416	0.1042	0.0405	na	0.014
Rev_Coup	0.0308	0.0260	0.0381	0.1236	0.1365	0.1005	0.0317	0.077	na
Publ_Edu_pct	0.0304	0.0223	0.0340	0.2998	0.3734	0.2713	0.0318	0.157	0.099
Area	0.0286	0.0243	0.0325	0.1392	0.1521	0.1131	0.0268	na	na

Table 4: PIPs obtained for FLS datasets in different packages and simulation scenarios

5.2 Results for birthwt dataset

Variable	res_BACE1	res_BACE2	res_BACE3	res_ParMA	bicreg	iBMA.bicreg
age	0.4340	0.5024	0.4237	0.3807	0.257	0.265
lwt	0.8113	0.7932	0.8346	0.7152	0.665	0.656
Drace_2	0.4460	0.5181	0.5146	0.5192	0.129	0.133
Drace_3	0.3068	0.4287	0.3591	0.5491	0.032	0.032
Dsmoke_1	0.4258	0.5404	0.4913	0.6068	0.141	0.146
Dptl_1	0.9903	0.9891	0.9901	0.9875	1	1
Dptl_2	0.1797	0.2705	0.2146	0.1943	0.014	na
Dht_1	0.8120	0.8105	0.8507	0.7508	0.633	0.622
Dui_1	0.4392	0.5257	0.5036	0.5282	0.223	0.229
ftv	0.1653	0.2670	0.1937	0.1814	0.015	na
Dptl_3	na	na	na	na	0.028	0.029

Table 5: PIPs obtained for birthwt datasets in different packages and simulation scenarios

References

Albis MLF, Mapa DS. 2014. Bayesian Averaging of Classical Estimates in Asymmetric Vector Autoregressive (AVAR) Models. MPRA Paper 55902, University Library of Munich, Germany.
URL https://ideas.repec.org/p/pra/mprapa/55902.html

Atwood CL. 1980. Estimators for the truncated beta-binomial distribution .
URL https://www.osti.gov/biblio/5359803

Białowolski P, Kuszewski T, Witkowski B. 2014. Bayesian averaging of classical estimates in forecasting macroeconomic indicators with application of business survey data. *Empirica* **41**: 53–68.
URL https://ideas.repec.org/a/kap/empiri/v41y2014i1p53-68.html

Doppelhofer G, Weeks M. 2006. Jointness of Growth Determinants. Cambridge Working Papers in Economics 0542, Faculty of Economics, University of Cambridge.

Doppelhofer G, Weeks M. 2009. Jointness of Growth Determinants. *Journal of Applied Econometrics* **24**: 209–244.

Greene W. 1999. *Econometric Analysis*. Upper Saddle River: Prentice Hall, 4th edition.

Hofmarcher P, Crespo Cuaresma J, Grün B, Humer S, Moser M. 2018. Bivariate jointness measures in Bayesian Model Averaging: Solving the conundrum. *Journal of Macroeconomics* **57**: 150–165. ISSN 0164-0704.
URL https://www.sciencedirect.com/science/article/pii/S0164070418300314

Johansen S. 1995. *Likelihood-Based Inference in Cointegrated Vector Autoregressive Models*. Oxford University Press.

Kass RE, Raftery AE. 1995. Bayes Factors. *Journal of the American Statistical Association* **90**: 773–795. ISSN 0162-1459.

Leamer E. 1978. *Specification Searches*. New Jersey, USA: John Wiley & Sons. ISBN 0471015202.

Ley E, Steel M. 2007. Jointness in Bayesian Variable Selection with Applications to Growth Regression. *Journal of Macroeconomics* **29**: 476–493.

Mitchell TJ, Beauchamp JJ. 1988. Bayesian Variable Selection in Linear Regression. *Journal of the American Statistical Association* **83**: 1023–1032.
URL https://www.tandfonline.com/doi/abs/10.1080/01621459.1988.10478694

Tripathi RC, Gupta RC, Gurland J. 1994. Estimation of parameters in the beta binomial model. *Annals of the Institute of Statistical Mathematics* **46**: 317–331. ISSN 0020-3157.
URL http://link.springer.com/10.1007/BF01720588

van Dijk D. 2004. Forecasting US Inflation Using Model Averaging. Econometric Society 2004 Australasian Meetings 143, Econometric Society.
URL https://ideas.repec.org/p/ecm/ausm04/143.html

Xavier Sala-i-Martin and Gernot Doppelhofer and Ronald I Miller. 2004. Determinants of Long-Term Growth: A Bayesian Averaging of Classical Estimates (BACE) Approach. *American Economic Review* **94**: 813–835.
URL https://ideas.repec.org/a/aea/aecrev/v94y2004i4p813-835.html